

Ellic's Exercise Camp: Engaging Children in Physical Activity Through Virtual Reality Gaming

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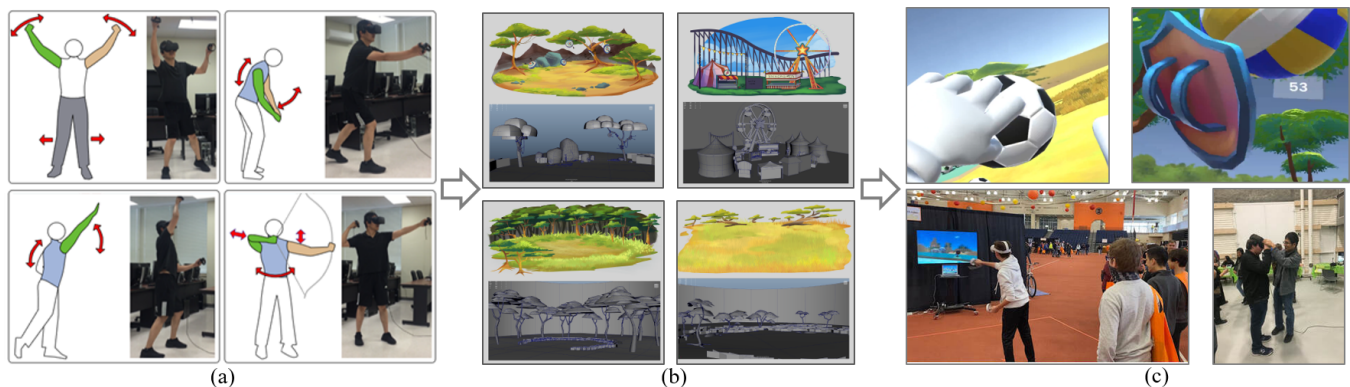


Figure 1: An overview of our virtual reality exercise game, featuring (a) the design of sports activities and interactive elements, (b) game design and asset creation, and (c) gameplay highlights and photographs from exhibitions.

ABSTRACT

Incorporating virtual reality (VR) games into children's physical activity routines offers an innovative way to promote exercise in a fun and engaging manner. By immersing children in captivating digital worlds that require physical movement to navigate, VR games transform daily sports activities into exciting adventures, as shown in Figure 1. Set within a vibrant cartoon environment, this game features a virtual exercise camp that captivates and draws children in, keeping them engaged and enthusiastic about participating in physical activities. We have crafted several mini-games inspired by various sports activities, each designed to be both fun and easy to learn. Adorable cartoon characters come to life to guide players through the gameplay, enhancing the interactive experience. This immersive approach entertains and encourages children to embrace exercise as a joyful part of their daily routine.

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1 GAME DESIGN

1.1 Mini-Game based on Sports Activities

This game is a collection of mini-games, all the mini-games are designed based on sports activities. The current version of the game contains six mini-games, as shown in Figure 2:

- Defense Castle: Pirates lay siege to the castle, and players valiantly wield their shields to deflect incoming cannonballs. This game draws inspiration from ball catching and blocking activities.
- Goalkeeper: Players take on the role of a goalkeeper who stop the NPC from scoring by reacting to balls that vary in speed, angle, and curve.
- Archery: Players test their precision by shooting at moving targets, mirroring the actions and skills of real-life archery.
- Volleyball: Players need to bounce the volleyball into baskets, enhancing their precision skills and spatial awareness.



Figure 2: The screenshots display the collection of mini-games based on sports activities.



Figure 3: The design of our main menu, emulating an exercise camp experience.

- **Skeet:** The NPC tosses skeets skyward, and players need to shoot them before they land on the ground.
- **Lucky Ducky:** Inspired by classic theme park games, players aim a water gun at moving wooden ducks to score points.

Each mini-game is designed to last one minute, providing quick and engaging gameplay sessions. We offer three difficulty levels for all mini-games, allowing users to select the level that best matches their skill and comfort. This ensures a tailored gaming experience for players from newcomers to seasoned gamers.

1.2 Game Environment

The game adopts a cartoon art style suitable for children. To optimize performance on VR devices, reduce motion sickness, and achieve smooth physical simulations, all game models feature a low polygon count.

Set up as an exercise camp, the game mirrors the essence of summer camps familiar to children, as shown in Figure 3. Players



Figure 4: The design of Ellic, a cartoonish elephant character who interacts with the players in the game.

navigate a map to choose mini-games, earning badges for achievements and progress. This structure encourages exploration and active participation, offering an experience similar to real-life summer camp adventures and rewards. Each mini-game features crafted environments inspired by diverse landscape features like deserts, wetlands, grassland, and forests.

1.3 NPC and Tutorial

Ellic, a cartoonish elephant character, is designed to accompany players throughout the game, offering guidance and enhancing the gameplay experience, as shown in Figure 4. This feature is particularly beneficial for children, offering them a friendly and engaging companion that makes learning and exploring within the game more enjoyable and intuitive.

Tutorials for all the mini-games are enriched with illustrations and animations. Given that these mini-games are inspired by everyday sports events, players can quickly grasp the gameplay mechanics.

2 GAME SETUP

This game is compatible with Oculus Rift and Oculus Quest devices and supports both wired and wireless connections. As outlined by Meta, using the Quest requires users to be at least 10 years old (may vary by country). Our game adheres to the same age requirement and includes safeguards for adult supervision.

3 USER STUDIES AND EXHIBITIONS

The previous version of this game was showcased at Frameless Symposium [Cao et al. 2020]. We have conducted a formal user study on engagement and exercise intensity [Cao et al. 2021]. In addition, this game has been adopted in research on exploring the benefits of VR games for older adults [Fan et al. 2022].

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